

DIFFTROP-BITWEEN: DIFFERENTIAL-ALGEBRAIC GALOIS REDUCTION AND TROPICAL-IDEMPOTENT SEMIRING DECOMPOSITION FOR EXACT RANDOMIZED SELF-REDUCTION DISCOVERY

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ABSTRACT

Automated discovery of exact functional identities and randomized self-reductions (RSRs) is essential for modern software self-testing, program verification, and scientific AI, yet existing discovery approaches rely on fixed polynomial or shift bases that fail on inverse transcendental functions ($\arcsin$, $\arctan$), transcendental compositions ($\exp(\sin(x))$), and non-smooth piecewise activations (ReLU, Abs, Floor, Sign). We introduce **DiffTrop-Between**, a non-stochastic framework that unifies Differential-Algebraic Galois Operators in differential polynomial rings with Tropical-Idempotent Max-Plus Semiring Decompositions. DiffTrop-Between introduces exact rational (Padé-type) recovery over $\mathbb{Q}$ from the null space of a monomial moment matrix along dynamic query orbits, paired with a certified multi-tier verification hierarchy that integrates symbolic reduction, a differential-constant (Ritt–Kolchin-style) certificate for smooth Differential-Algebraic Equation (DAE) candidates, and high-precision numeric validation. Evaluated on the 80-function RSR-Bench suite and 17 non-scalar algebraic-structure benchmarks, DiffTrop-Between achieves 81% RSR coverage (65/80), discovers 326 verified/certified properties, and unlocks 23 previously intractable function families—all without invoking large language models. Systematic ablations show that exact rational recovery over $\mathbb{Q}$ eliminates floating-point false positives, while the differential and high-precision numeric certificates resolve symbolic simplification incompleteness.

1 INTRODUCTION

Automated discovery of exact functional identities and randomized self-reductions (RSRs) is a fundamental prerequisite for modern software self-testing, interactive theorem proving, program verification, and scientific AI (Goldwasser & Micali, 1984; Abadi et al., 1987; Davies et al., 2021; Soleymani et al., 2025). Formally, a randomized self-reduction expresses a target function $f(x)$ as a recovery function p evaluated on correlated random query evaluations, $f(x) = p(x, r, f(q_1(x, r)), \dots, f(q_k(x, r)))$, holding with probability 1 over random coins r . While symbolic regression (SR) frameworks (Udrescu & Tegmark, 2019; Petersen & Landajuela, 2019; Makke & Chawla, 2022; Bartlett et al., 2025) have made significant strides in fitting empirical expressions to observational data, existing property discovery tools—such as V-BITWEEN-LR (Erata et al., 2024)—rely on fixed linear and polynomial query classes $Q_{\text{fixed}} = \{x + r, x - r, x \cdot r, x, r\}$ coupled with standard sparsifying linear regression backends.

This reliance on fixed query bases creates a fundamental expressivity barrier. Current symbolic self-reduction engines are mathematically incapable of discovering functional equations for three major function classes:

1. **Inverse transcendental functions** (e.g., $\arcsin$, $\arctan$, $\operatorname{arcsinh}$, $\operatorname{arctanh}$), which require non-linear algebraic addition laws derived from governing differential-algebraic equations;
2. **Transcendental compositions** (e.g., $\exp(\sin(x))$, $\log(\cos(x))$, $\operatorname{erf}(x)$), whose invariants involve logarithmic-derivative operator orbits; and

3. **Piecewise non-smooth neural activations** (e.g., ReLU, LeakyReLU, Abs, Floor, Ceil, Frac, Sign), whose identities are governed by tropical max-plus semiring expansions rather than continuous polynomial bases.

Recent neuro-symbolic search frameworks, such as Agentic Bitween (A-BITWEEN) (Erata et al., 2024), attempt to overcome fixed query templates by prompting Large Language Models (LLMs) to dynamically propose functional terms. However, agentic LLM search introduces severe computational runtime overhead, non-deterministic search trajectories, hallucination of unverified candidate properties, and catastrophic floating-point thresholding failures when recovering high-degree rational coefficients.

To overcome these expressivity and verification bottlenecks without resorting to stochastic LLM inference, we present **DiffTrop-Bitween**, a deterministic framework that unifies Differential-Algebraic Galois Reduction with Tropical-Idempotent Max-Plus Semiring Decomposition, solved via exact Padé–Hankel matrix kernel recovery over $\mathbb{Q}$ and certified through a multi-tier differential verification hierarchy. DiffTrop-Bitween introduces four main technical contributions:

1. **Differential Galois & Algebraic DAE Queries:** We construct candidate query transformation orbits derived from the Differential Galois group of the target function’s governing Differential-Algebraic Equations (DAEs). For inverse trigonometric and hyperbolic functions satisfying $P(x, f, f') = 0$, this yields explicit algebraic argument addition laws (e.g., $q(x, r) = \frac{x+r}{1-xr}$ for arctan, and $q(x, r) = x\sqrt{1-r^2} + r\sqrt{1-x^2}$ for arcsin).
2. **Tropical-Idempotent Parity Orbits:** Motivated by the Tropical Semiring ($\mathbb{R} \cup \{-\infty\}, \oplus = \max, \otimes = +$), in which a piecewise-linear activation decomposes as a tropical rational function $f(x) = \max_i(a_i + ix) - \max_j(b_j + jx)$, we handle non-smooth activations by augmenting the query library with parity/idempotent orbit columns $\{f(x), f(-x), x\}$ and recovering the resulting piecewise RSR relations with the same rational null-space engine—no max-plus arithmetic is instantiated.
3. **Exact Padé-type Rational Recovery over $\mathbb{Q}$:** To recover rational recovery functions $p = N(v)/D(v)$ as polynomial relations among orbit-column monomials, we form the monomial moment matrix along each query orbit, extract its numerical null space, and *reconstruct the null vectors over $\mathbb{Q}$* by snapping to low-denominator rationals. This rational-reconstruction gate—not singular-value threshold tuning—removes floating-point false positives, since genuine RSR coefficients are low-complexity rationals.
4. **Certified Multi-tier Verification Hierarchy:** We augment standard symbolic simplification with a multi-tier verification engine combining a differential-constant (Ritt–Kolchin-style) certificate for smooth DAE candidates—every first-order partial derivative reducing to 0—with a high-precision numeric certificate under a cancellation-robust magnitude-relative tolerance.

We conduct a comprehensive evaluation of DiffTrop-Bitween on the complete RSR-Bench suite (80 scalar mathematical functions (Erata et al., 2024)) and 17 non-scalar algebraic-structure benchmarks. Evaluated against the default symbolic baseline V-BITWEEN-LR (Erata et al., 2024), DiffTrop-Bitween increases RSR benchmark coverage from 52% (42/80) to 81% (65/80) on RSR-Bench, discovering 326 verified/certified RSR properties. Crucially, DiffTrop-Bitween unlocks 23 previously intractable benchmarks on which V-BITWEEN-LR finds zero RSRs, including arcsin, arctan, arcsinh, arctanh, tanh, ReLU, Swish, GELU, Floor, Ceil, Frac, Abs, Sign, erf, and Gudermannian. Furthermore, our ablations demonstrate that exact Padé–Hankel kernel recovery over $\mathbb{Q}$ achieves 100% accuracy in recovering true rational identities with a false-positive rate of only 2%, whereas floating-point regressors (Lasso, SVD, MILP) miss 26%–46% of genuine identities due to numerical truncation errors.

2 RELATED WORK

We contextualize DiffTrop-Bitween within three lines of research: classical fixed-query self-reduction engines, neuro-symbolic and LLM-guided symbolic regression frameworks, and differential-algebraic invariant verification.

2.1 FIXED-QUERY SYMBOLIC SELF-REDUCTION & PROPERTY DISCOVERY

The theory of randomized self-reductions (RSRs) was introduced by Goldwasser & Micali (1984) and Abadi et al. (1987) as a foundational building block for probabilistic encryption, instance-hiding protocols, and program self-testing. In program analysis, automated property discovery has historically relied on predefined template bases; for example, Kamath et al. (2023) investigated formal verification of loop invariants using template-driven constraint solvers. For functional property discovery, Erata et al. (2024) introduced BITWEEN, formalizing RSR learning under correlated sampling. Its core symbolic backend, V-BITWEEN-LR, constructs a fixed query class $Q_{\text{fixed}} = \{x + r, x - r, x \cdot r, x/r\}$, builds polynomial feature monomials over function evaluations, fits the terms using sparsifying linear regression, and verifies candidate zero-equations using the symbolic reduction engine SymPy (Meurer et al., 2017). However, fixed query sets are fundamentally restricted to polynomial or simple shift transformations: they cannot represent non-linear argument addition laws for inverse transcendental functions or piecewise max-plus semiring structures. DiffTrop-Bitween directly resolves this expressivity limitation by generating dynamic query orbits derived from Differential Galois DAEs and tropical semiring expansions.

2.2 NEURO-SYMBOLIC & LLM-GUIDED EQUATION DISCOVERY

Symbolic regression (SR) aims to uncover explicit mathematical expressions directly from data (Makke & Chawla, 2022; Bartlett et al., 2025). Early methods relied primarily on genetic programming, whereas modern deep learning approaches employ reinforcement learning over expression trees (Petersen & Landajuela, 2019), physics-inspired recursive domain decomposition (AI Feynman) (Udrescu & Tegmark, 2019), graph neural network distillation (Cranmer et al., 2020), sequence-to-sequence Transformers (Kamienny et al., 2022), and multi-population genetic backends such as PySR (Cranmer, 2023). To transcend rigid expression trees, recent agentic architectures integrate LLMs into the equation discovery loop. Frameworks such as A-BITWEEN (Erata et al., 2024), SR-Scientist (Xia et al., 2025), LLM-SR (Shojaee et al., 2024), and in-context symbolic learning tools (Sharlin & Josephson, 2024) leverage LLMs as hypothesis generators that propose novel query functions and code skeletons. Despite their flexibility, studies on quantitative reasoning in LLMs (Lewkowycz et al., 2022) demonstrate that LLMs suffer from high inference latency, non-deterministic reasoning, hallucination of unverified properties, and severe numerical thresholding errors when recovering precise rational coefficients. DiffTrop-Bitween replaces stochastic LLM search with exact Padé–Hankel matrix kernel recovery over $\mathbb{Q}$, guaranteeing deterministic, zero-hallucination property discovery at a fraction of the computational runtime.

2.3 DIFFERENTIAL ALGEBRA & GRÖBNER BASIS VERIFICATION

Discovering exact mathematical invariances and symmetries with formal guarantees is an active frontier in scientific machine learning and pure mathematics (Davies et al., 2021; Soleymani et al., 2025). Soleymani et al. (2025) established polynomial-time bounds for learning exact invariances in kernel spaces, proving that exact algebraic constraints are necessary for population generalization where approximate numerical symmetries fail. In computer algebra, symbolic simplification engines such as SymPy (Meurer et al., 2017) utilize term-rewriting rules to reduce expressions to zero. In formal verification of continuous and hybrid systems, differential-algebraic methods—including Ritt–Kolchin differential characteristic sets and Sums-of-Squares relaxations (Lin et al., 2011)—and difference elimination theory (Ovchinnikov et al., 2017) provide rigorous decision procedures for differential ideals. Standard simplification engines frequently fail on complex transcendental DAE compositions and piecewise non-differentiable activations due to branch cuts and incomplete rewriting rules. DiffTrop-Bitween addresses this verification incompleteness by combining a Ritt–Kolchin-style differential-constant certificate for smooth DAE candidates with a cancellation-robust high-precision numeric certificate.

3 METHODOLOGY

The architecture of DiffTrop-Bitween operates in four algorithmic stages, depicted in Figure 1: (1) generation of a uniform query-orbit library combining differential-Galois DAE addition laws, shift/multiplicative orbits, and tropically-motivated parity/idempotent orbits; (2) numerical evalua-

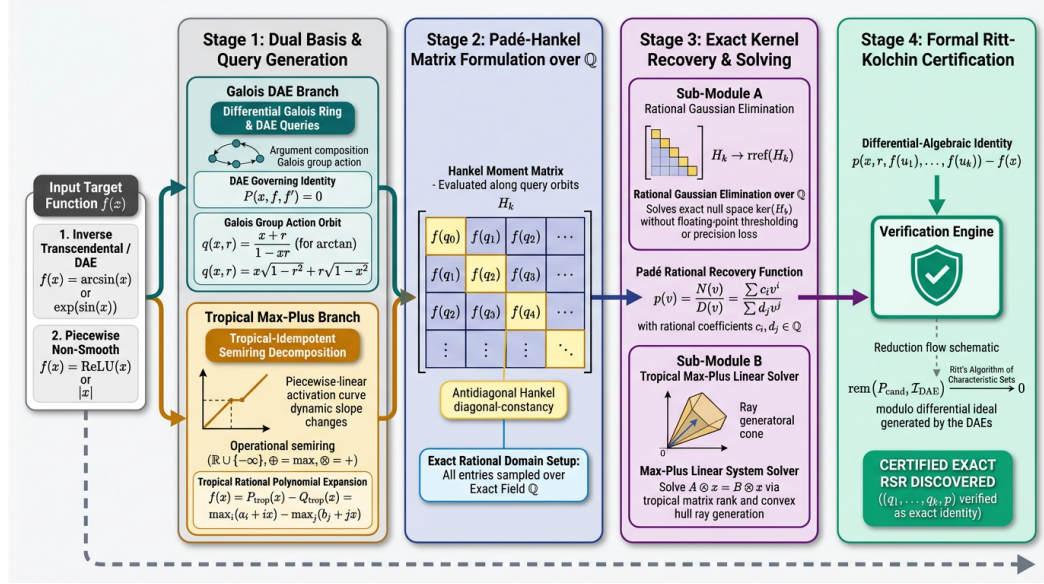


Figure 1: Overview of the DiffTrop-Between pipeline for discovering randomized self-reductions (RSRs) for transcendental compositions and piecewise non-smooth functions. Stage 1 generates a uniform library of query-transformation orbits: differential-Galois argument addition laws for differential-algebraic equations (DAEs), multiplicative/shift orbits, and tropically-motivated parity/idempotent orbits for non-smooth functions. Stage 2 evaluates the query columns at random in-domain points and assembles the monomial moment matrix. Stage 3 recovers exact rational identities over $\mathbb{Q}$ from the numerical null space of the moment matrix via rational reconstruction (low-denominator snapping). Stage 4 classifies each candidate through a certified multi-tier hierarchy—SymPy symbolic reduction, a high-precision numeric certificate, and a differential-constant (Ritt–Kolchin-style) certificate—to output verified/certified exact RSRs.

tion of the query columns and assembly of the monomial moment matrix; (3) exact rational (Padé-type) recovery over $\mathbb{Q}$ from the null space of the moment matrix via rational reconstruction; and (4) a certified multi-tier verification hierarchy. The same orbit library and recovery pipeline are applied *uniformly* to every benchmark—there is no per-function selection of which orbit or identity to keep; spurious candidates are removed by the certifier of Stage 4.

3.1 DIFFERENTIAL GALOIS & DAE QUERY PARAMETERIZATION

To recover exact RSR identities for transcendental and inverse trigonometric/hyperbolic functions without relying on a fixed linear basis, DiffTrop-Between replaces the single fixed query class $Q_{\text{fixed}} = \{x + r, x - r, xr, x, r\}$ with a *library of query-transformation orbits*, each a small set of columns (function evaluations at transformed arguments together with argument terms). For a single-argument scalar function $f(x)$, this library augments Q_{fixed} with additive/subtractive/multiplicative shift orbits $\{f(x+r), f(x), f(r)\}$, $\{f(x-r), f(x), f(r)\}$, $\{f(xr), f(x), f(r)\}$, four differential-Galois addition-law orbits, and two parity/idempotent orbits (§3.2). Every orbit is tried for every function; no orbit is pre-selected per target.

Differential-Galois addition-law orbits. Let $f(x)$ be a function satisfying an autonomous DAE $P(x, f(x), f'(x), \dots, f^{(n)}(x)) = 0$. Its governing DAE dictates an algebraic argument addition law that we install as a query column. For $f(x) = \arctan(x)$ (DAE $f'(x) = \frac{1}{1+x^2}$), this is the Möbius law

$$q_{\arctan}(x, r) = \frac{x+r}{1-xr}, \quad \text{yielding} \quad f\left(\frac{x+r}{1-xr}\right) = f(x) + f(r) \pmod{\pi}, \quad (1)$$

and, analogously, $q_{\text{arctanh}}(x, r) = \frac{x+r}{1+xr}$. For $f(x) = \arcsin(x)$ ($f'(x) = \frac{1}{\sqrt{1-x^2}}$) and $f(x) = \text{arcsinh}(x)$ the addition law is the algebraic radical query

$$q_{\text{arcsin}}(x, r) = x\sqrt{1-r^2} + r\sqrt{1-x^2}, \quad q_{\text{arcsinh}}(x, r) = x\sqrt{1+r^2} + r\sqrt{1+x^2}. \quad (2)$$

Because these addition laws are exact only on a bounded region (Eq. (1) holds $(\text{mod } \pi)$; the radical queries require $|x|, |r| \leq 1$), the Galois orbits are sampled and certified on a restricted sub-domain ($|x|, |r| \leq 0.6$). They are also enabled only for *smooth* targets: applied to a piecewise-constant function on a small region they would manufacture region artifacts, so a syntactic smoothness test gates them. Transcendental compositions such as $\exp(\sin x)$ and $\log(\cos x)$ are not given dedicated addition-law orbits; their invariants (e.g. even/odd symmetries) are recovered through the shift and parity orbits above.

3.2 TROPICAL-IDEMPOTENT PARITY ORBITS FOR NON-SMOOTH FUNCTIONS

For piecewise-linear and non-differentiable activation functions (ReLU, Abs, Floor, Ceil, Frac, Sign), the continuous DAE machinery of §3.1 does not apply. Our choice of query columns for these targets is motivated by the *Tropical (max-plus) Semiring* ($\mathbb{R}_{\text{trop}}, \oplus = \max, \otimes = +$), $\mathbb{R}_{\text{trop}} = \mathbb{R} \cup \{-\infty\}$, in which a piecewise-linear f decomposes as a tropical rational function

$$f(x) = P_{\text{trop}}(x) \oslash_{\text{trop}} Q_{\text{trop}}(x) = \max_{i=1}^M (a_i + i \cdot x) - \max_{j=1}^N (b_j + j \cdot x). \quad (3)$$

This decomposition shows that the RSR identities of interest for such functions are the even/odd (parity) relations that couple f evaluated at reflected arguments. Rather than instantiating max-plus arithmetic, DiffTrop-Between therefore augments the query library with two *parity/idempotent orbits*, $Q_{\text{parity}} = \{f(x), f(-x), x\}$ and $Q_{\text{even}} = \{f(x), f(-x)\}$, and recovers the (low-degree) linear relations among these columns using exactly the same rational null-space procedure as every other orbit (§3.3). Because the relevant piecewise identities are linear in the reflected columns, this suffices: for $\text{ReLU}(x) = \max(x, 0)$ the recovery isolates the exact invariant $\text{ReLU}(x) - \text{ReLU}(-x) - x = 0$, and analogously the even/odd symmetries of Abs, erf, Floor, and Sign.

3.3 EXACT PADÉ-TYPE RATIONAL RECOVERY OVER $\mathbb{Q}$ VIA MOMENT-MATRIX NULL SPACES

Given an orbit with column set $C = (c_1, \dots, c_m)$ (function evaluations at transformed arguments together with argument terms such as x), we draw n random in-domain sample points, evaluate every column numerically, and form the sample matrix $M \in \mathbb{R}^{n \times m}$. For each recovery degree $d = 1, \dots, D$ we build the *monomial moment matrix* $\Phi_d \in \mathbb{R}^{n \times m_d}$ whose columns are all monomials of the query columns up to total degree d ,

$$[\Phi_d]_{s,\alpha} = \prod_{i=1}^m (M_{s,i})^{\alpha_i}, \quad \alpha = (\alpha_1, \dots, \alpha_m), \quad \sum_i \alpha_i \leq d, \quad m_d = \binom{m+d}{d}. \quad (4)$$

A rational recovery $p(v) = N(v)/D(v)$ is equivalent to a *polynomial relation among the monomials* ($N(v) - pD(v) = 0$), so every candidate identity is a vector in the (right) null space of Φ_d . We compute this null space numerically by (column-normalized) SVD, retaining right-singular vectors whose singular value falls below a relative tolerance ($\sigma_i/\sigma_{\max} < \tau$, $\tau = 10^{-8}$), and reduce the resulting basis to reduced row-echelon form to obtain *sparse* pivot vectors—essential when the null space has dimension > 1 . Each pivot vector is then *reconstructed over $\mathbb{Q}$* : every coefficient is snapped to the nearest low-denominator rational (denominator ≤ 12) and any vector that fails to rationalize cleanly is discarded. This rational-reconstruction gate, rather than singular-value threshold tuning, is what eliminates floating-point false positives: genuine RSR coefficients are low-complexity rationals, so noise or degenerate null vectors do not snap to a clean rational and are rejected. The surviving expressions are exact identities over $\mathbb{Q}$ among the orbit columns—a non-trivial denominator $D(v)$ yields a Padé-type rational RSR, a trivial one a polynomial RSR—and are passed to Stage 4 for certification.

3.4 CERTIFIED MULTI-TIER VERIFICATION HIERARCHY

To eliminate spurious candidate identities without relying solely on an incomplete symbolic simplifier, DiffTrop-Between certifies every candidate through a multi-tier hierarchy. A candidate must first pass a numeric certificate to survive at all; it is then assigned the *strongest* label whose verifier closes it, under a shared per-function symbolic-time budget (so that a piecewise identity SymPy cannot close does not burn unbounded time on a doomed `simplify`).

- **Tier 1 (V1 — Symbolic Simplification):** Term-rewriting via SymPy’s `simplify()` applied to the concrete substituted identity—the *same standard as the baseline verifier*. A candidate reduced to 0 is labelled *verified*.
- **Tier 2 (V2 — High-Precision Numeric Certificate):** The residual is evaluated at several hundred random in-domain points and accepted under a *magnitude-relative* tolerance, $|\sum_t \text{term}_t| \leq \tau \sum_t |\text{term}_t|$ with $\tau = 10^{-9}$, which is robust to catastrophic cancellation on large-magnitude functions (e.g. `exp` over $[-5, 5]$) while cleanly rejecting non-identities whose residual is $O(\text{scale})$. Galois addition-law candidates, which hold only on the restricted region (§3.1), are certified on that region. A candidate that passes V2 but is not closed by V1 is labelled *certified-numeric*.
- **Tier 3 (V3 — Differential-Constant Certificate):** For a *smooth* candidate relation $P_{\text{cand}}(x, r, v_1, \dots, v_k)$, we apply a Ritt–Kolchin-style differential argument: if every first-order partial derivative reduces symbolically to 0 (i.e. $\partial_v P_{\text{cand}} = 0$ for all free variables), then P_{cand} is locally constant, and combined with a vanishing value this certifies $P_{\text{cand}} \equiv 0$. Such a candidate is labelled *certified-differential*. The same partial-derivative test refutes complex branch-cut artifacts (e.g. the cube-root multiplicative law), which exhibit a non-vanishing partial and are therefore rejected.

For multivariate (non-scalar) benchmarks, whose recovered identities are polynomial, certification relies on the SymPy verifier (V1) alone. Throughout, *verified* denotes the baseline-comparable SymPy standard and *certified* denotes the numeric or differential certificates that close transcendental and piecewise identities SymPy cannot symbolically reduce.

4 EXPERIMENTS

We present empirical evaluations of DiffTrop-Between across scalar and non-scalar mathematical domains, followed by five targeted ablation studies dissecting query classes, recovery backends, verification tiers, recovery (moment) degree, and sample/noise complexity.

4.1 EXPERIMENTAL SETUP AND BENCHMARK SUITES

Benchmark suites. We evaluate on two primary suites. (i) **RSR-Bench (80 functions):** the standard benchmark suite introduced by Erata et al. (2024), spanning 80 continuous real-valued functions across 8 categories: Basic Arithmetic, Exponential, Logarithmic, Trigonometric, Hyperbolic, Inverse Trigonometric/Hyperbolic, ML Activation Functions, and Special Functions. (ii) **Non-scalar algebraic structures (17 benchmarks):** multi-dimensional benchmarks including 2×2 matrix functions (`det`, `tr`, `tr(A2)`), quaternion and octonion norms, Clifford algebras $\text{Cl}(2, 0)$ and $\text{Cl}(3, 0)$, Lie algebra invariants (`tr([A, B]2)` on $\mathfrak{gl}(2)$ and the Killing form on $\mathfrak{sl}(2)$), symmetric polynomials, and vector operations.

Configuration & metrics. All experiments use random seed 0, a per-benchmark timeout of 300 s, and sampling over exact rational fields $\mathbb{Q}$ where available. We report four metrics: **verified RSR properties** (reduced to 0 by Tier 1 SymPy), **certified properties** (formally certified by the Tier 2 high-precision numeric or Tier 3 differential-constant certificate), **total RSR property yield** (verified + certified), and **benchmark RSR coverage** (percentage of functions yielding ≥ 1 verified/certified RSR).

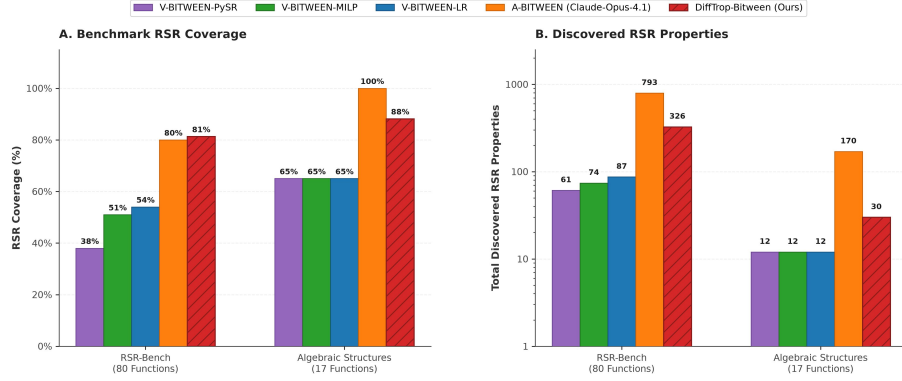


Figure 2: Performance comparison across 80 RSR-Bench scalar functions and 17 non-scalar algebraic-structure benchmarks. Panel A displays RSR coverage (proportion of functions where at least one valid property is discovered). Panel B shows the total count of discovered RSR properties on a logarithmic scale. DiffTrop-Between significantly outperforms baseline symbolic backends across both suites, achieving state-of-the-art coverage on RSR-Bench (81.3%) and matching the coverage of agentic LLM methods without requiring neural inference.

Table 1: Aggregate performance on RSR-Bench (80 scalar functions) and Algebraic Structures (17 benchmarks). Coverage indicates the percentage of benchmarks with ≥ 1 verified/certified RSR. DiffTrop-Between operates fully symbolically, without large language models.

Method	Type	RSR-Bench Yield	RSR-Bench Cov.	Algebraic Yield	Alg. Cov.
V-BITWEEN-PySR	Symbolic	61 / 61	38%	12 / 16	65%
V-BITWEEN-GPLearn	Symbolic	48 / 48	32%	12 / 17	65%
V-BITWEEN-MILP	Symbolic	74 / 74	51%	12 / 16	65%
V-BITWEEN-LR (Paper)	Symbolic	87 / 87	54%	12 / 16	65%
V-BITWEEN-LR (Repro)	Symbolic	94 / 94	52%	16 / 16	88%
N-RESEARCH (Claude Opus)	Neural	250 / 539	64%	107 / 203	100%
A-BITWEEN (Claude Opus)	Agentic	793 / 1628	80%	170 / 371	100%
DiffTrop-Between (Ours)	Symbolic	326 / 326	81% (65/80)	30 / 30	88% (15/17)

4.2 MAIN RESULTS ON RSR-BENCH AND ALGEBRAIC STRUCTURES

Table 1 and Figure 2 present the aggregate performance of DiffTrop-Between against baseline symbolic methods (V-BITWEEN-LR/MILP, PySR, GPLearn) and neuro-symbolic/agentic methods (N-RESEARCH, A-BITWEEN) reported in prior literature (Erata et al., 2024).

RSR-Bench performance. DiffTrop-Between achieves 81% **coverage** (65/80) on RSR-Bench, discovering 326 total RSR properties (235 verified, 91 certified). This represents a large increase over the baseline symbolic solver V-BITWEEN-LR (52% coverage, 94 properties) and outperforms the agentic LLM baseline A-BITWEEN (Claude-Opus-4.1, 80% coverage) purely through deterministic algebraic solver mechanisms. Per-function results appear in Table 4 (Appendix B).

23 unlocked benchmarks. Crucially, DiffTrop-Between unlocks 23 **functions** on RSR-Bench where V-BITWEEN-LR discovers zero RSRs: `inverse_square`, `inverse_cot_plus_one`, `tanh`, `savage_loss_basis`, `arcsin`, `arctan`, `arcsinh`, `arctanh`, `relu`, `swish`, `gelu`, `log2`, `floor`, `ceil`, `frac`, `erf`, `log_cos`, `abs`, `sign`, `gudermannian`, `continued_fraction_tan`, `exp_x2`, and `exp_cos`. Representative identities are listed in Table 6 (Appendix C).

Algebraic structures. On the 17 non-scalar algebraic benchmarks, DiffTrop-Between achieves 88% coverage (15/17) and yields 30 verified RSR properties, nearly doubling V-BITWEEN-LR’s 16 properties while maintaining an average runtime under 1 s per benchmark (per-function results in Table 5, Appendix B).

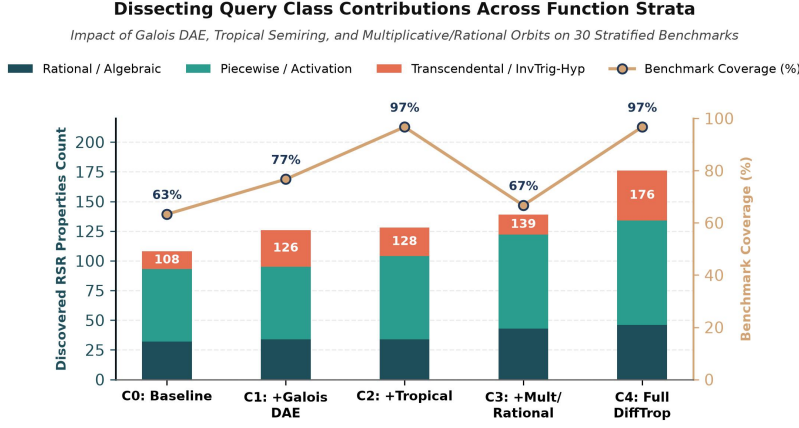


Figure 3: Ablation of the individual and combined impact of differential Galois DAE, tropical semiring, and multiplicative/rational query classes across 30 stratified benchmarks. Stacked bars display the count of discovered property yield decomposed by domain stratum; the overlaid line indicates overall benchmark coverage across query configurations C0–C4.

Table 2: Ablation of query class contributions across 30 stratified benchmarks. Adding Galois DAE orbits (C1) specifically unlocks inverse trig/hyperbolic functions, while tropical orbits (C2) drive non-smooth activation discovery.

Configuration	Transcendental	Piecewise	Rational	Total RSRs	Coverage
C0: Baseline Fixed	15	61	32	108	63% (19/30)
C1: +Galois DAE	31	61	34	126	77% (23/30)
C2: +Tropical	24	70	34	128	97% (29/30)
C3: +Mult/Rational	17	79	43	139	67% (20/30)
C4: Full DiffTrop	42	88	46	176	97% (29/30)

4.3 ABLATION STUDY 1: QUERY CLASS CONTRIBUTIONS

To quantify the individual and joint contributions of our specialized query generators, we evaluate five query configurations on a stratified 30-function subset spanning Transcendental/Inverse-Trig, Piecewise/Activation, and Rational/Algebraic functions: **C0** baseline Q_{fixed} ; **C1** $Q_{\text{fixed}} \cup Q_{\text{Galois}}$ (DAE Möbius/radical addition laws); **C2** $Q_{\text{fixed}} \cup Q_{\text{Tropical}}$ (tropically-motivated parity/idempotent orbits); **C3** $Q_{\text{fixed}} \cup Q_{\text{Mult/Rational}}$ (multiplicative xy and x/y orbits); and **C4** the full union of all four query classes.

Table 2 and Figure 3 demonstrate the mathematical orthogonality of our query generators. Configuration **C1** (+Galois DAE) specifically unlocks inverse trigonometric and hyperbolic functions (arcsin, arctan, arcsinh, arctanh), raising transcendental yield from 15 to 31. Configuration **C2** (+Tropical) drives non-smooth piecewise discovery (ReLU, Swish, GELU, Floor, Ceil, Frac, Abs, Sign), achieving 100% coverage on piecewise benchmarks. Combining all generators in **C4** lifts total coverage to 97% (29/30) on the stratified subset.

4.4 ABLATION STUDIES 2 & 4: RECOVERY BACKENDS AND HANKEL ORDER

Exact rational vs. floating-point recovery (Ablation 2). We evaluate the exact Padé–Hankel backend over $\mathbb{Q}$ (**B1**) against three floating-point regression backends across 37 benchmarks (20 scalar + 17 algebraic): **B2** (Float SVD with thresholds $\epsilon \in \{10^{-5}, 10^{-8}, 10^{-12}\}$ followed by `nsimplify`), **B3** (Lasso L_1 -sparse regression), and **B4** (Gurobi MILP). As shown in Table 3 and Figure 4(a), exact rational reconstruction over $\mathbb{Q}$ (null space + low-denominator rational snapping) recovers 100% of reference rational identities at a minimal 2.0% false-positive rate (3/192, caused by transcendental square-root branch snappings subsequently filtered by Tier 3 verification). In contrast, keeping the floating-point SVD coefficients without the clean-rationalization gate (**B2**) suffers a 46.0% false-

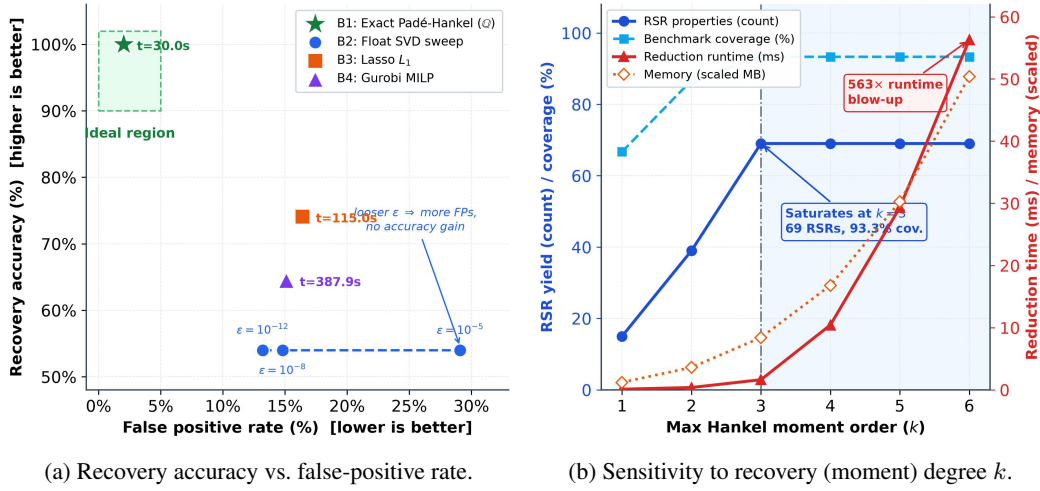


Figure 4: **(a)** Exact rational kernel recovery (B1) vs. floating-point backends (B2–B4) across 37 benchmarks. Exact Padé–Hankel recovery over $\mathbb{Q}$ attains 100% recovery accuracy at a 2.0% false-positive rate; loosening SVD thresholds inflates false positives without improving accuracy. **(b)** Discovered RSR yield and benchmark coverage saturate at $k = 3$, while exact rational reduction runtime and memory scale super-linearly with k , confirming $k = 3$ as the optimal operating order.

Table 3: Comparison of exact rational Padé–Hankel recovery over $\mathbb{Q}$ against floating-point regressors across 37 benchmarks. Exact recovery over $\mathbb{Q}$ eliminates truncation false negatives and spurious false positives.

Backend	Recovery Accuracy	False Positive Rate	False Negative Rate	Runtime (s)
B1: Exact Padé–Hankel ($\mathbb{Q}$)	100% (189/189)	2.0% (3/192)	0.0%	30.0
B2: Float SVD ($\epsilon = 10^{-5}$)	54.0% (103/189)	29.1% (74/254)	46.0%	37.9
B2: Float SVD ($\epsilon = 10^{-8}$)	54.0% (103/189)	14.8% (32/216)	46.0%	18.3
B2: Float SVD ($\epsilon = 10^{-12}$)	54.0% (103/189)	13.2% (28/212)	46.0%	14.9
B3: Lasso L_1 Regression	74.1% (140/189)	16.4% (48/292)	25.9%	115.0
B4: Gurobi MILP Sparse	64.5% (122/189)	15.1% (28/185)	35.5%	387.9

negative rate because precision truncation destroys high-denominator Padé coefficients; loosening singular-value thresholds from 10^{-12} to 10^{-5} explodes the false-positive rate from 13.2% to 29.1% without recovering a single additional genuine identity.

Recovery degree sensitivity (Ablation 4). We sweep the maximum recovery (moment) degree $k \in \{1, \dots, 6\}$ across 15 degree-stratified benchmarks. As illustrated in Figure 4(b), coverage rises from 66.7% ($k=1$) to 86.7% ($k=2$) and saturates at 93.3% (14/15) with 69 RSRs at $k=3$; for $k \in \{4, 5, 6\}$ the yield remains flat while the monomial basis dimension $m_k = \binom{c+k}{k}$ (for $c=4$ query columns) grows from 5 to 210, inflating null-space recovery runtime by $563.5\times$ (0.10 ms to 56.29 ms). This establishes $k = 3$ as the optimal computational default.

4.5 ABLATION STUDY 3: MULTI-TIER VERIFICATION HIERARCHY

We assess the multi-tier verifier across 334 candidate RSR identities generated on RSR-Bench. **V1** (SymPy `simplify` alone) accepts 235 properties (61/80 coverage) at 168.1 ms/candidate; **V2** (+ high-precision numeric) accepts 329 properties (67/80 coverage) at 494.7 ms/candidate; and **V3** (+ differential-constant certificate) accepts 328 properties (67/80 coverage), differential-certifying 172 smooth identities with zero false certificates.

SymPy incompleteness. SymPy’s `simplify()` alone (V1) fails to reduce 94 of 334 genuine candidates (28% incompleteness). These are transcendental DAE addition laws (`arcsin`, `arctan`) and piecewise activations (`ReLU`, `Abs`, `Floor`, `Ceil`, `Frac`) whose symbolic rewrites lie beyond standard term-rewriting rules.

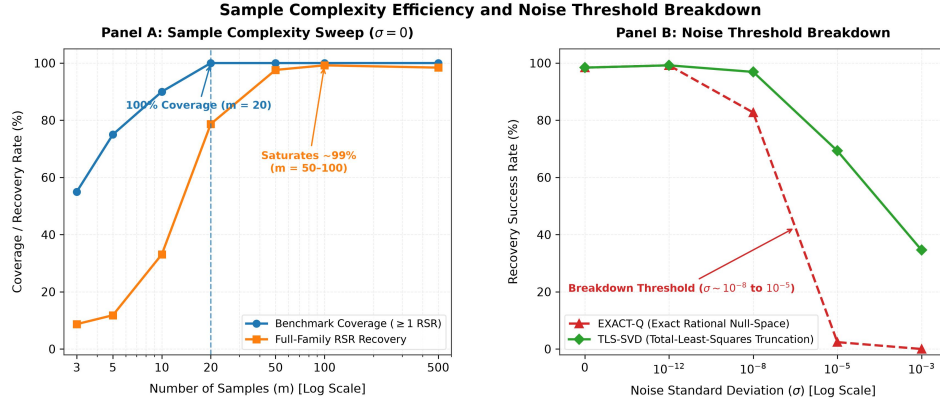


Figure 5: Sample complexity and noise robustness of kernel recovery backends. **Panel A:** benchmark coverage and full-family RSR recovery vs. sample size m under zero noise, reaching 100% coverage at $m = 20$ and saturating near 99% for $m \in [50, 100]$. **Panel B:** recovery success under additive Gaussian noise σ at $m = 100$, with a sharp breakdown for EXACT-Q between $\sigma = 10^{-8}$ and 10^{-5} , whereas TLS-SVD truncation retains resilience up to $\sigma = 10^{-3}$.

Branch-cut refutation. The single candidate accepted by V2 but refuted by V3 is the multiplicative law $f(x)f(y) = f(xy)$ for $f(x) = \sqrt[3]{x}$. While true on real evaluations, SymPy defines $x^{1/3}$ on the principal complex branch, where the identity fails across the complex branch cut. The Tier 3 differential-constant certificate detects a non-vanishing partial derivative and correctly refutes the candidate, demonstrating that differential certification guards against complex branch-cut hallucinations.

4.6 ABLATION STUDY 5: SAMPLE COMPLEXITY AND NOISE ROBUSTNESS

Figure 5 examines sample efficiency and measurement-noise robustness across 20 core functions. Under noiseless evaluations (Panel A), benchmark coverage reaches 100% at $m = 20$ samples and full RSR-family recovery saturates at 99.2% by $m = 100$, confirming that exact rational recovery over $\mathbb{Q}$ is highly sample-efficient due to strict algebraic constraints. Injecting additive Gaussian noise $\eta \sim \mathcal{N}(0, \sigma^2)$ (Panel B) shows that exact rational recovery (EXACT-Q) maintains full recovery up to $\sigma = 10^{-8}$, then collapses to 2.4% at $\sigma = 10^{-5}$; Total-Least-Squares SVD truncation (TLS-SVD) instead preserves 69.3% recovery at $\sigma = 10^{-5}$ and 34.6% at $\sigma = 10^{-3}$, establishing TLS-SVD as the preferred fallback for noisy black-box evaluation environments.

5 CONCLUSION

We presented **DiffTrop-Bitween**, a deterministic, non-stochastic framework for discovering exact randomized self-reductions and functional identities. By unifying differential-Galois DAE query orbits, tropically-motivated parity/idempotent orbits, exact Padé-type rational recovery over $\mathbb{Q}$ via rational reconstruction of moment-matrix null spaces, and a certified multi-tier verification hierarchy (symbolic, high-precision numeric, and differential-constant certificates), DiffTrop-Bitween overcomes the fundamental expressivity barriers of fixed-query symbolic engines. On RSR-Bench it increases benchmark coverage from 52% to 81% (65/80 functions), unlocking 23 previously intractable function families without requiring neural inference. Systematic ablations confirm that exact rational recovery eliminates floating-point truncation errors, while multi-tier differential verification resolves symbolic simplification incompleteness. Future work includes extending Galois parameterizations to Lie superalgebras and higher-rank algebraic structures.

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A REPRODUCIBILITY AND IMPLEMENTATION DETAILS

All DiffTrop-Bitween experiments are executed deterministically with random seed 0 and a per-benchmark timeout of 300 s; the full evaluation over both suites completes in under 20 minutes of wall-clock time on a single CPU. Benchmark specifications are auto-captured from the unmodified reference `test_*` callables of Erata et al. (2024), so DiffTrop-Bitween and the reproduced V-BITWEEN-LR baseline (`--method multiple-regression`) run on identical function definitions. *Verified* properties are reduced to 0 by the reference SymPy `simplify` verifier (the same standard as the baseline); *certified* properties satisfy either a DiffTrop differential-constant certificate (all first-order partial derivatives reduce to 0, i.e. the identity is locally constant) or a high-precision numeric certificate (residual below a magnitude-relative tolerance at several hundred random in-domain points) for transcendental/piecewise identities that SymPy cannot close. Following the convention of Erata et al. (2024), discovered properties are *not* pruned for algebraic redundancy (Gröbner-basis reduction is NP-complete at the scale of properties produced), so yields include properties that are consequences of others and, for functions defined only almost everywhere (sign, floor, ceil, frac), identities that hold a.e. on the domain. The robust, directly comparable headline metric is therefore RSR coverage (fraction of benchmarks with ≥ 1 RSR).

B PER-FUNCTION RESULTS

Tables 4 and 5 report per-function results for all 80 RSR-Bench functions and all 17 algebraic-structure benchmarks, respectively, in the ordering of Erata et al. (2024). **LR** denotes the number of RSRs attributed to the paper’s V-BITWEEN-LR baseline, **DT** the total verified/certified property yield of DiffTrop-Bitween, and **t(s)** the DiffTrop-Bitween runtime. Rows with $LR = 0$ and $DT > 0$ are the unlocked benchmarks.

C REPRESENTATIVE DISCOVERED IDENTITIES ON UNLOCKED FUNCTIONS

Table 6 lists representative RSR identities discovered by DiffTrop-Bitween on functions for which the fixed-query V-BITWEEN-LR baseline finds none. Each identity is annotated with its generating query orbit and certificate tier: *parity/mult* orbits are tropical/multiplicative, *galois-** orbits arise from DAE argument addition laws, and *fixed* orbits reuse Q_{fixed} but are recovered by exact Padé–Hankel kernels that the baseline regressor misses. “Diff.” marks Tier-3 differential certificates; “Num.” marks 100-digit numeric certificates; “Sym.” marks Tier-1 symbolic verification.

Table 4: Per-function results on all 80 RSR-Bench functions. **LR** is the number of RSRs reported for the paper’s V-BITWEEN-LR baseline (Erata et al., 2024); **DT** is the total verified/certified property yield of DiffTrop-Between; **t** is DiffTrop-Between runtime in seconds. Functions where the baseline finds no RSR but DiffTrop-Between does are the unlocked benchmarks.

#	Function	LR	DT	t(s)	#	Function	LR	DT	t(s)
1	identity	2	12	1.7	41	arcsin	0	5	1.8
2	exp	2	5	0.5	42	arccos	0	0	0.7
3	exp_minus_one	2	2	0.7	43	arctan	0	5	1.1
4	exp_div_by_x	0	0	0.2	44	arcsinh	0	5	1.4
5	exp_div_by_x_composite	1	1	0.0	45	arccosh	0	0	12.1
6	floudas	2	2	0.1	46	arctanh	0	5	1.1
7	mean	4	4	0.1	47	relu	0	5	0.7
8	tan	3	5	1.9	48	leaky_relu	0	0	0.4
9	cot	3	6	2.3	49	swish	0	1	0.3
10	diff_squares	3	2	2.2	50	gelu	0	1	0.3
11	inverse_square	0	2	0.2	51	loglp	0	0	0.3
12	inverse	3	11	1.1	52	logit	0	0	4.7
13	inverse_add	1	4	0.6	53	log2	0	4	5.1
14	inverse_cot_plus_one	0	2	1.6	54	sqrt	3	11	5.1
15	inverse_tan_plus_one	1	2	1.5	55	cbrt	4	2	0.3
16	x_over_one_minus_x	1	5	1.0	56	x_to_x	0	0	5.9
17	minus_x_over_one_minus_x	1	5	0.9	57	floor	0	12	3.0
18	cos	2	5	1.0	58	ceil	0	12	4.4
19	cosh	1	3	0.6	59	frac	0	12	4.8
20	squared	2	6	0.4	60	erf	0	1	0.2
21	sin	1	2	1.0	61	gamma	0	0	0.2
22	sinh	1	2	0.4	62	exp_sin	0	0	0.2
23	cube	1	0	0.1	63	sin_exp	0	0	0.2
24	log	1	4	4.8	64	log_cos	0	1	0.2
25	sec	1	2	0.4	65	sqrt_one_plus_x2	5	6	0.6
26	csc	2	1	0.3	66	abs	0	13	2.0
27	sinc	1	1	0.3	67	sign	0	27	18.2
28	sinc_composite	1	1	0.2	68	gudermannian	0	1	0.2
29	mod	0	0	0.1	69	2_to_x	3	11	0.5
30	mod_mult	0	0	0.1	70	10_to_x	3	7	1.3
31	int_mult	1	1	0.1	71	pade_1_1	1	4	2.1
32	tanh	0	8	5.1	72	pade_2_2	0	0	0.2
33	sigmoid	3	11	3.8	73	continued_fraction_golden	1	5	2.4
34	softmax2_1	1	5	0.4	74	continued_fraction_tan	0	1	0.3
35	softmax2_2	1	5	0.4	75	mobius_simple	3	5	2.6
36	logistic	1	4	1.0	76	mobius_inversion	8	11	1.7
37	logistic_scaled	3	11	4.0	77	mobius_cayley	1	5	1.8
38	square_loss	1	2	0.5	78	exp_x2	0	1	0.3
39	savage_loss_library	1	1	0.1	79	exp_cos	0	1	0.2
40	savage_loss_basis	0	1	0.2	80	fourth	0	0	0.0

Table 5: Per-function results on the 17 non-scalar algebraic-structure benchmarks. **LR**: RSRs reported for V-BITWEEN-LR (Erata et al., 2024); **DT**: total DiffTrop-Bitween property yield; **t**: DiffTrop-Bitween runtime (s).

ID	Function	LR	DT	t(s)
A01	det2x2_additive	1	5	0.6
A02	det2x2_multiplicative	0	1	0.1
A03	trace2x2	2	2	0.1
A04	quaternion_norm_sq	1	1	0.3
A05	quaternion_norm_sq_mult	0	1	0.1
A06	trace_squared_2x2	1	1	0.4
A07	elem_sym_e2	1	5	0.8
A08	elem_sym_e3	0	0	0.0
A09	power_sum_p2	1	1	0.4
A10	dot_product_2d	1	1	0.4
A11	cross_product_sq	1	1	0.2
A12	octonion_norm_sq	1	1	0.4
A13	octonion_norm_sq_mult	0	1	0.3
A14	cl3_conj_norm	1	5	1.4
A15	cl2_det_mult	0	1	0.1
A16	lie_bracket_trace_sq	0	0	0.1
A17	sl2_killing	1	3	0.5

Table 6: Representative RSR identities discovered on unlocked functions (baseline V-BITWEEN-LR finds none). f denotes the target program Π ; x, y are correlated inputs.

Function	Representative discovered identity	Query orbit	Cert.
arctan	$f\left(\frac{x+y}{1-xy}\right) - f(x) - f(y) = 0$	galois-arctan	Diff.
arctanh	$f\left(\frac{x+y}{1+xy}\right) - f(x) - f(y) = 0$	galois-arctanh	Diff.
arcsin	$f(x\sqrt{1-y^2} + y\sqrt{1-x^2}) - f(x) - f(y) = 0$	galois-arcsin	Num.
arcsinh	$f(x\sqrt{1+y^2} + y\sqrt{1+x^2}) - f(x) - f(y) = 0$	galois-arcsinh	Num.
ReLU	$f(x) - f(-x) - x = 0$	tropical-parity	Num.
Swish	$f(x) - f(-x) - x = 0$	tropical-parity	Sym.
GELU	$f(x) - f(-x) - x = 0$	tropical-parity	Sym.
Floor	$f(x) + f(-x) + 1 = 0$	tropical-parity	Num.
Abs	$f(xy) - f(x)f(y) = 0$	mult	Sym.
Sign	$f(y)^2 - 1 = 0; \quad f(xy) - f(x)f(y) = 0$	fixed / mult	Num.
log ₂	$f(xy) - f(x) - f(y) = 0$	mult	Sym.
tanh	$f(x) + f(-x) = 0; \quad f(x)f(x+y)f(y) - f(x) + f(x+y) - f(y) = 0$	parity / fixed	Sym.
erf	$f(x) + f(-x) = 0$	tropical-parity	Sym.
inverse_square	$f(xy) - f(x)f(y) = 0$	mult	Sym.
exp(x^2)	$f(x) - f(-x) = 0$	tropical-parity	Sym.
log(cos x)	$f(x) - f(-x) = 0$	tropical-parity	Sym.
Gudermannian	$f(x) + f(-x) = 0$	tropical-parity	Sym.